Computer Vision
Lecture 11: Point Processing

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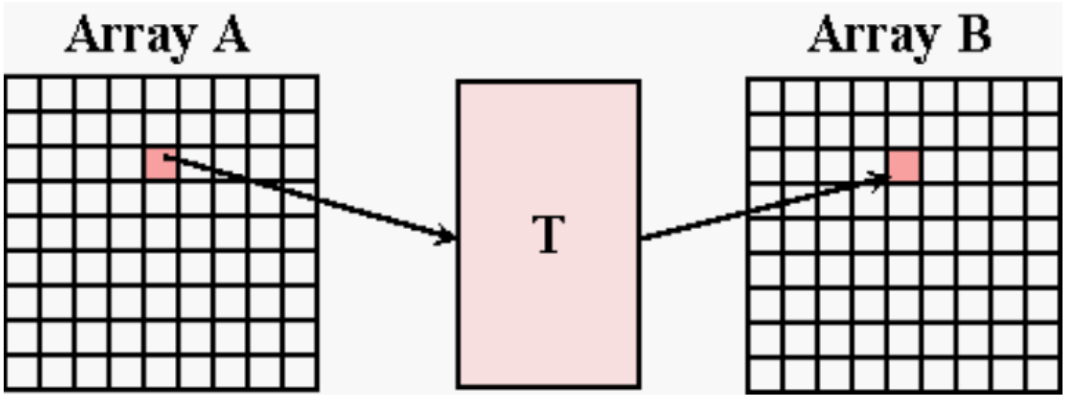
What is point processing?

- Point processing uses only the information in individual pixels to produce new images
- A transform(or filtering) may be computed on the basis of regional or global information and then applied to the individual points
- Applications to image enhancement are given for monochrome and color images.

Basic Point Processing

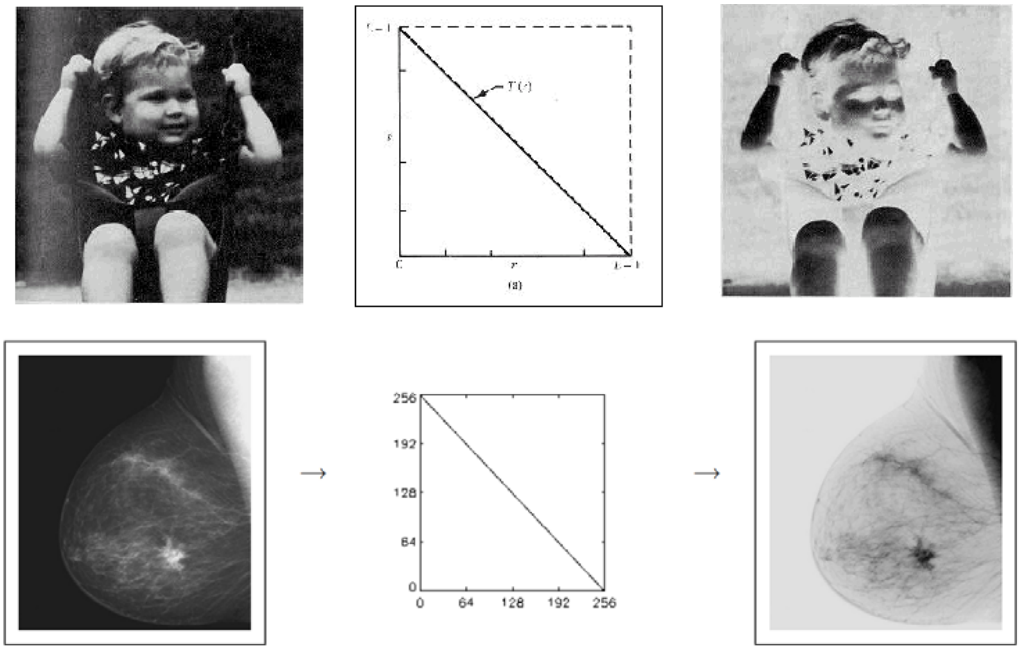
- Point processing is used to transform an image by operating on individual pixels. If array A represents an input image then an output array B is produced by a transformation

$$\mathbf{B}[x, y] = T\big[\mathbf{A}[x, y]\big]$$



Negative Image

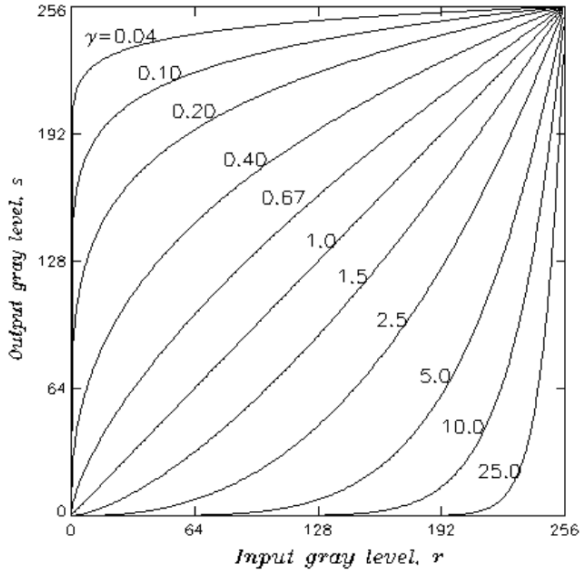
$$\mathbf{B}[x, y] = T\{\mathbf{A}[x, y]\} = 255 - \mathbf{A}[x, y]$$

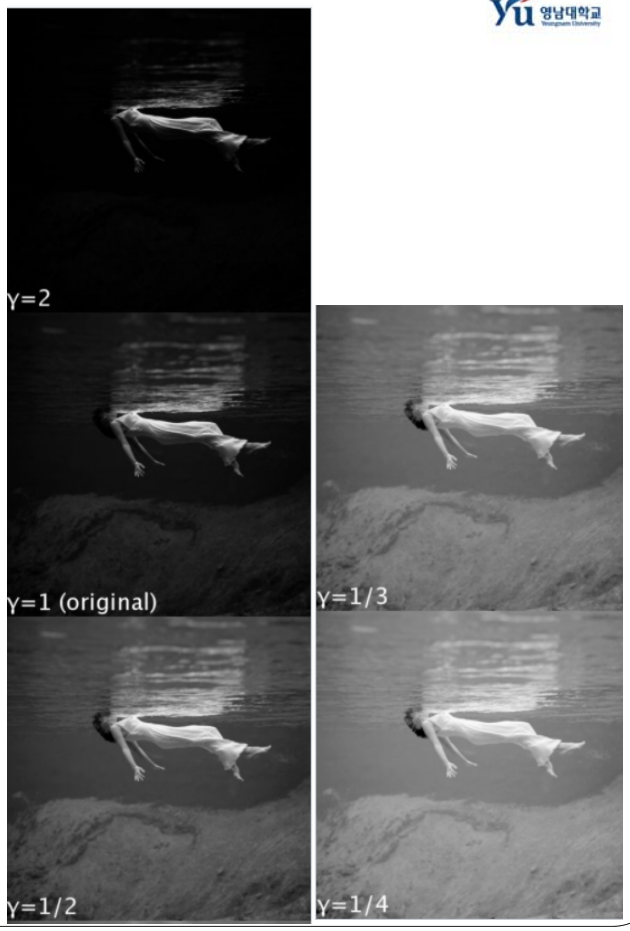


Gamma Correction

$$\mathbf{B}[x,y] = T\{\mathbf{A}[x,y]\} = (\mathbf{A}[x,y])^{\gamma}$$

Gamma Correction $s = r^{\gamma}$



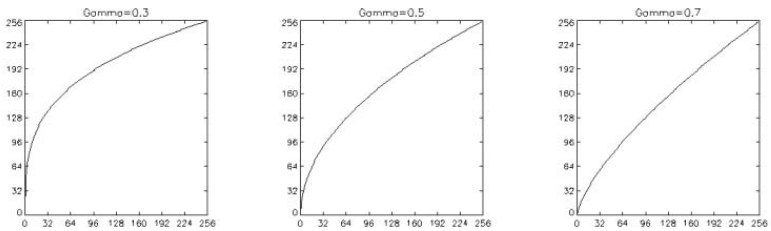


Gamma Correction

$$\mathbf{B}[x,y] = T\{\mathbf{A}[x,y]\} = (\mathbf{A}[x,y])^{\gamma}$$



Original $\gamma = 0.3$ $\gamma = 0.5$ $\gamma = 0.7$



Look-Up Table (LUT)

- **Lookup table** is an array that replaces runtime computation with a simpler array indexing operation

$$\mathbf{B}[x,y] = T\{\mathbf{A}[x,y]\} = (\mathbf{A}[x,y])^{\gamma}$$

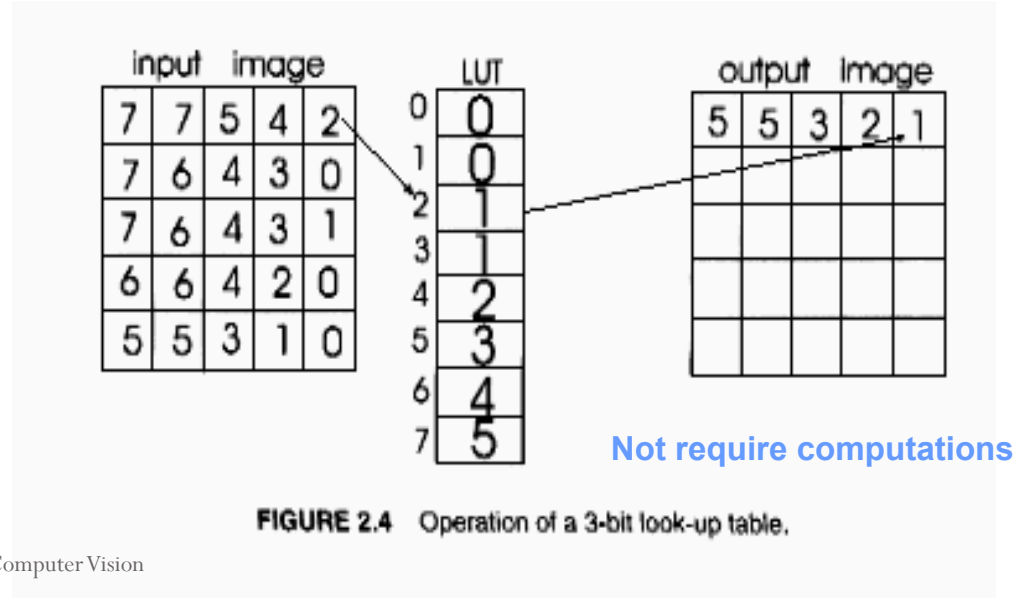
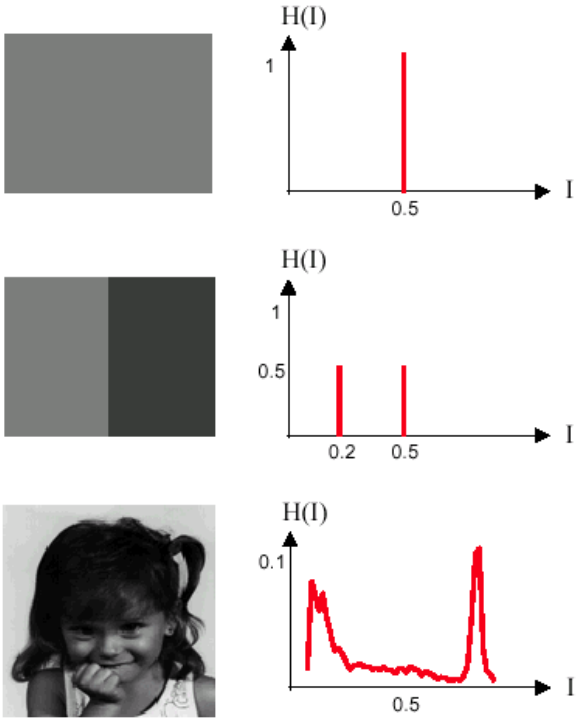


FIGURE 2.4 Operation of a 3-bit look-up table.

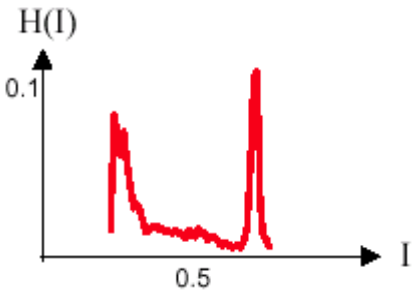
Histogram



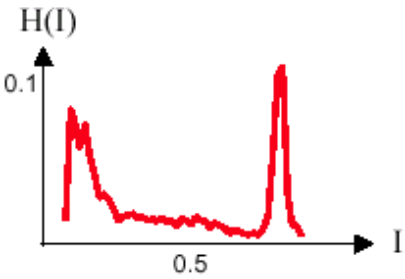

```
for(i=0,i<size, i++)
  for(j=0;j<size;j++)
      histo[image[i][j]]++;
```

Histogram









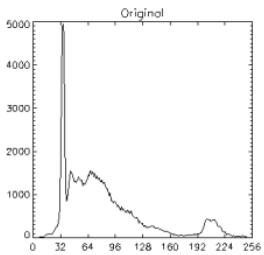
Effect of Gamma Correction on Histogram

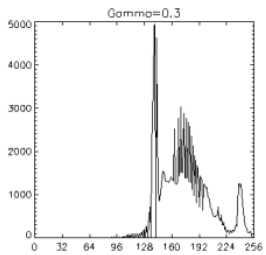


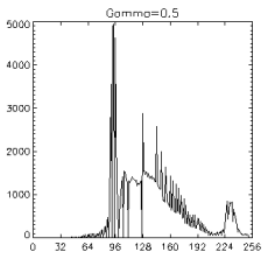


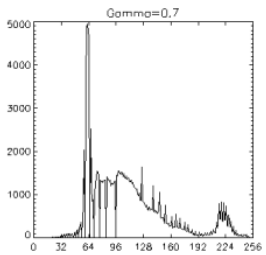








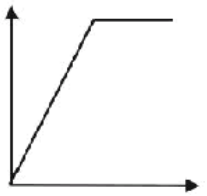




Histogram Equalization

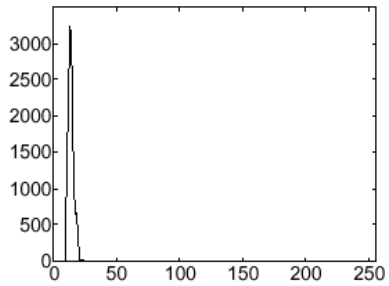
- Find a point transformation B = T[A] such that B has a uniform histogram

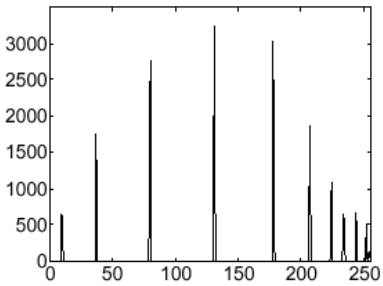




B = T[A]?







Transform of a Random Variable

X → Transform $f(\bullet)$ → $Y = f(X)$

$X = g(Y)$ ← Inverse Tr. $g(\bullet)$ ← Y

$$Y = f(X) = g^{-1}(X)$$

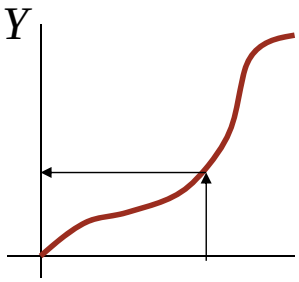
$$P_Y(Y \le y) = P_X(X \le x) = P_X(g(Y) \le x)$$

Differentiating with respect to y

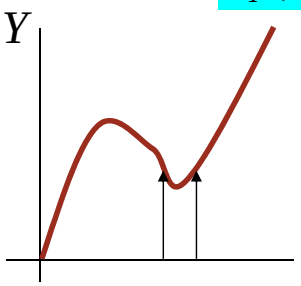
$$p_Y(y) = p_X(g(y))\left[\frac{d}{dy}g(y)\right]$$

One to one mapping

$$p_Y(y) = p_X(g(y))\left|\frac{d}{dy}g(y)\right|$$



One to one mapping : OK



One to multiple mapping : NO

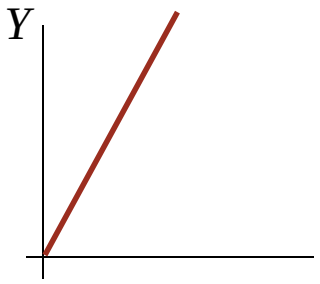
Transform of a Random Variable

❑ *Ex4.26; Find* $p_Y(y)$ *If* $Y = 2X$ *for given R.V.* X *with* $p_X(x)$

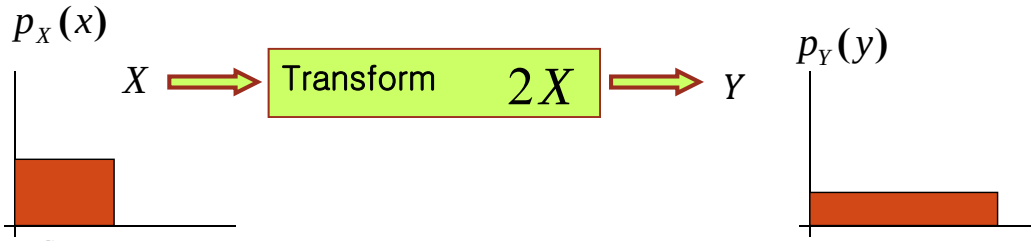
$$X = g(Y) \quad \Rightarrow \quad x = y/2$$

$$p_Y(y) = p_X(g(y))\left|\frac{d}{dy}g(y)\right|$$

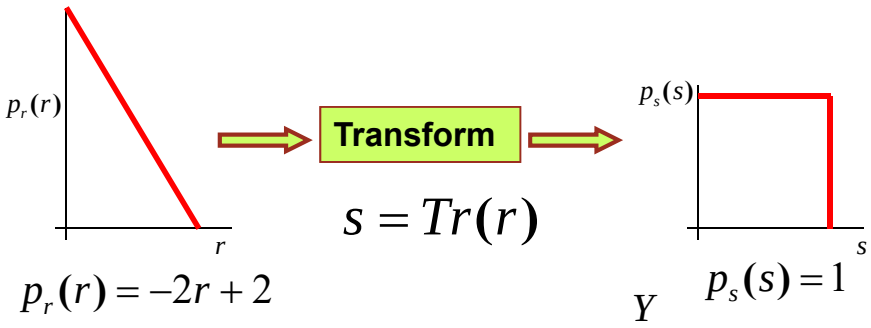
$$P_Y(y) = P_X\left(\frac{y}{2}\right)\left|\frac{d}{dy}\left(\frac{y}{2}\right)\right| = P_X\left(\frac{y}{2}\right)\left|\frac{1}{2}\right| = \frac{1}{2}P_X\left(\frac{y}{2}\right)$$



One to one mapping : OK

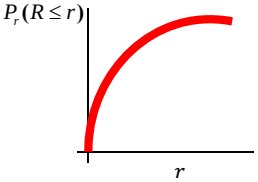


Transform of a Random Variable



$$p_r(r) = -2r + 2$$

$$p_s(s) = 1$$



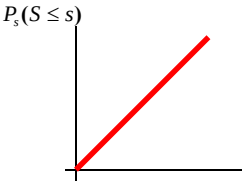
$$P_r(R \le r) = \int_0^r p_r(w)dw$$

$$= \int_0^r (-2w + 2)\,dw$$

$$= -r^2 + 2r$$

$$P_r(R \le r) = P_s(S \le s)$$

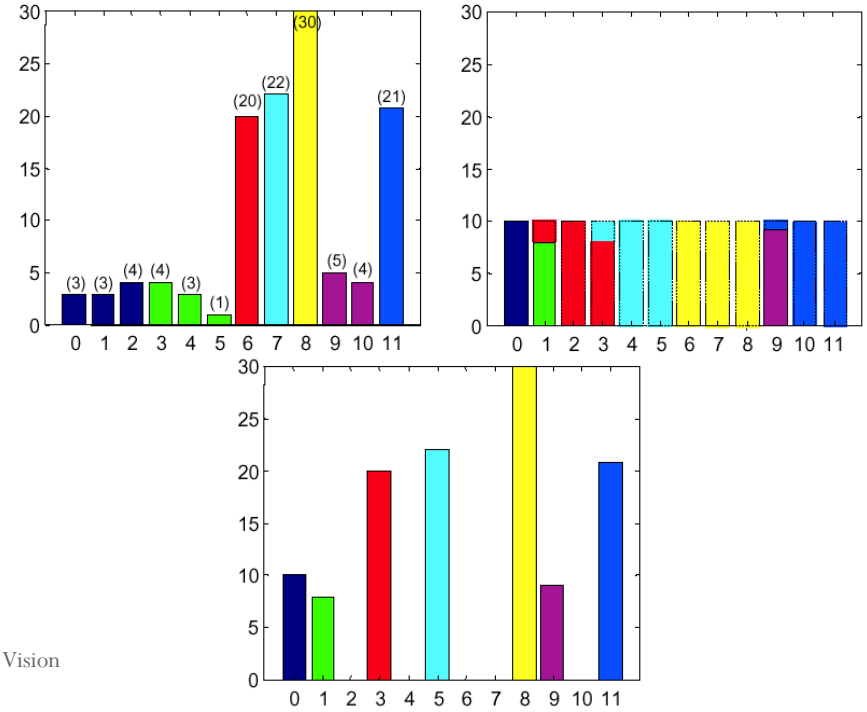
$$s = -r^2 + 2r$$



$$P_s(S \le s) = s$$

Histogram Equalization

- To obtain uniform histogram by spreading and shrinking



Histogram Equalization

- In OpenCV,

$$h(i) = round\left(\frac{cdf(i) - cdf_{min}}{N - cdf_{min}} \times L\right)$$

Where L is maximum pixel value, N is the number of pixels,

$cdf(i) = \sum_{j=0}^{i} n(j)$ is the cumulative distribution function ($n(j)$denotes the number of pixels with value i),

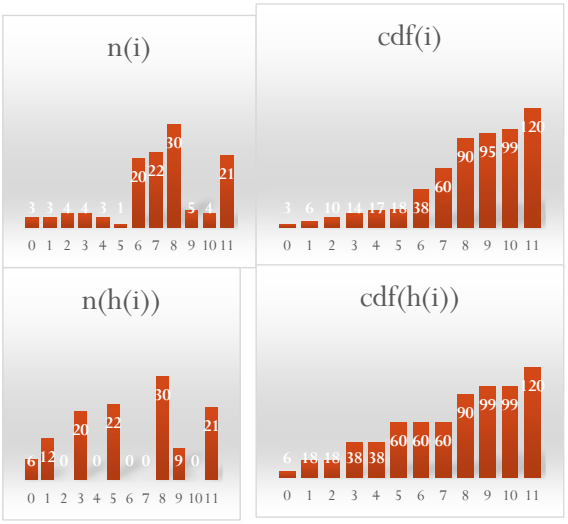
cdf_{min} is the cumulative distribution function (CDF) value of the smallest pixel value present in the image.

Histogram Equalization

In OpenCV, $h(i) = round\left(\frac{cdf(i)-cdf_{min}}{N-cdf_{min}} \times L\right)$, $cdf(i) = \sum_{j=0}^{i} n(j)$

$$L = 11, N = 120, cdf_{min} = cdf(0) = 3$$

i	n(i)	cdf(i)	h(i)	h(i)	n(h(i))	cdf(h(i))
0	3	3	0	0	6	6
1	3	6	0	1	12	18
2	4	10	1	2	0	18
3	4	14	1	3	20	38
4	3	17	1	4	0	38
5	1	18	1	5	22	60
6	20	38	3	6	0	60
7	22	60	5	7	0	60
8	30	90	8	8	30	90
9	5	95	9	9	9	99
10	4	99	9	10	0	99
11	21	120	11	11	21	120

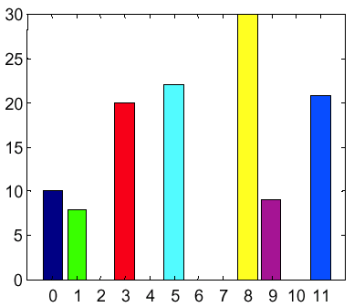


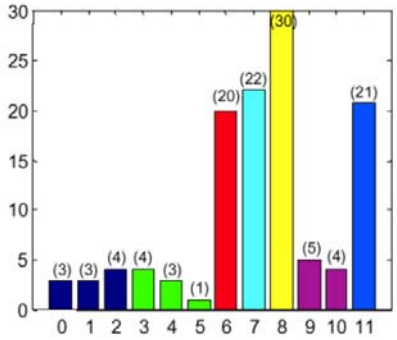
Histogram Equalization

- To obtain uniform histogram by spreading and shri

Method 1: $\left\lceil \frac{Maximum\ \ pixel\ \ value+1}{Total\ \ number\ \ of\ \ pixels} \times Sum - 1 \right\rceil$

Method 2: $round\left(\frac{Maximum\ \ pixel\ \ value}{Total\ \ number\ \ of\ \ pixels} \times Sum\right)$





			#method 1		#method2	
Brightness	number	sum of num.	normal sum	integer1	normal sum	interger2
0	3	3	-0.70	0	0.28	0
1	3	6	-0.40	0	0.55	1
2	4	10	0.00	0	0.92	1
3	4	14	0.40	1	1.28	1
4	3	17	0.70	1	1.56	2
5	1	18	0.80	1	1.65	2
6	20	38	2.80	3	3.48	3
7	22	60	5.00	5	5.50	6
8	30	90	8.00	8	8.25	8
9	5	95	8.50	9	8.71	9
10	4	99	8.90	9	9.08	9
11	21	120	11.00	11	11.00	11

Histogram Equalization

4	4	3	3
4	4	3	3
4	1	2	3
0	1	2	3

Input image

7	7	5	5
7	7	3	5
7	1	2	5
0	1	2	5

Histogram equalizes image

$$h(i) = round\left(\frac{cdf(i)-cdf_{min}}{N-cdf_{min}} \times L\right), cdf(i) = \sum_{j=0}^{i} n(j)$$

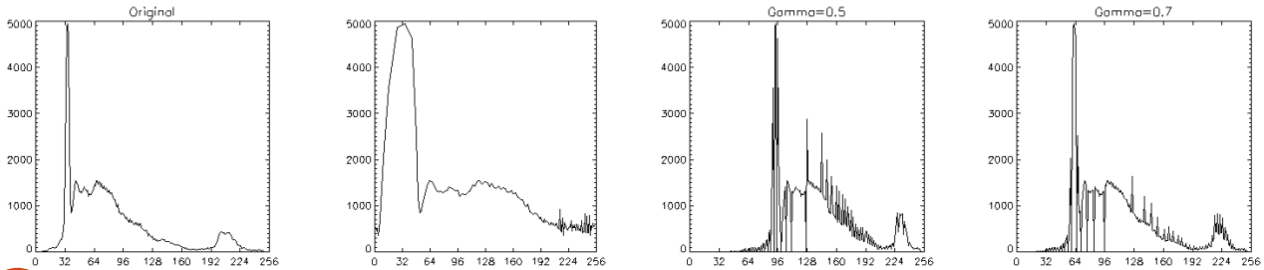
$$L = 7, N = 16,\ cdf_{min} = cdf(0) = 1$$

i	n(i)	cdf(i)	h(i)
0	1	1	0
1	2	3	1
2	2	5	2
3	6	11	5
4	5	16	7
5	0	16	7
6	0	16	7
7	0	16	7

Gamma Correction vs. Histogram Equalization

Original | Equalized | $\gamma = 0.5$ | $\gamma = 0.7$





Examples: Histogram Equalization of color image

- How to enhance a color image using Histogram Equalization?



http://www.ee.siue.edu/~sumbaug/CVIPbook_PPLec/chapter%208b.ppt

Examples: Histogram Equalization of color image

- One technique would be to equalize each of the color planes independently
- What artifacts could this create?



Red Channel Green Channel Blue Channel

Examples: Histogram Equalization of color image

- One technique would be to equalize each of the color planes independently



Examples: Histogram Equalization of color image

- One another technique would be to equalize the luminance while maintaining the hue and saturation values the same



Luminance (value)



Luminance Equalized

Examples: Histogram Equalization of color image

- One another technique would be to equalize the luminance while maintaining the hue and saturation values the same



Comparison of Color Histogram Equalization



Original



RGB Equalized Independently



Luminance Equalization



Single Band (R)



Single Band (G)



Single Band (B)

Image Subtraction

- Image subtraction can be used to suppress background information and highlight other information
- Applications: Medical imaging, motion detection, change detection







Image Subtraction

- Image subtraction can be used to suppress background information and highlight other information. The background image is taken first. Then a radiological contrast agent is injected into the bloodstream to form the second image. The difference image is on the right.

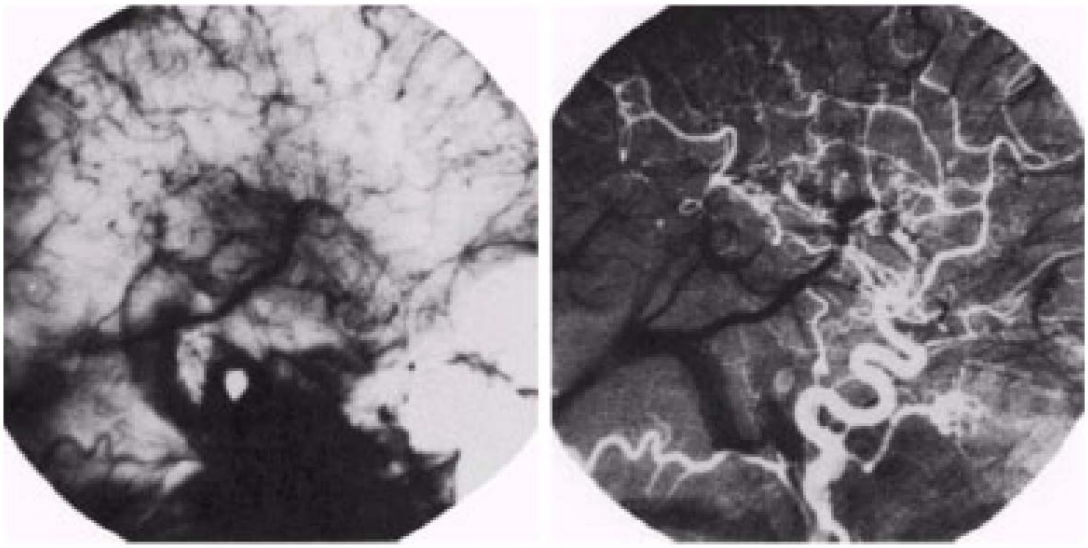


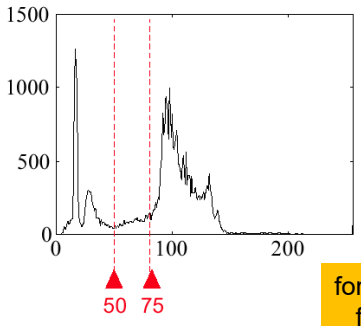
Image Thresholding

- **Thresholding** is the simplest method of segmenting image
- From a grayscale image, thresholding can be used to create binary image

Original

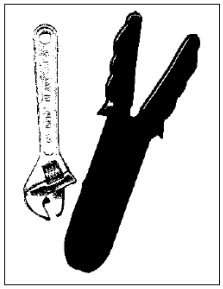


Histogram

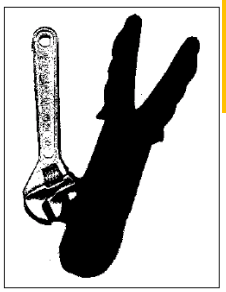



```
for(i=0,i<size, i++)
  for(j=0;j<size;j++)
     if (in_image[i][j]]>=50)
        out_image[i][j]=1;
     else  out_image[i][j]=0;
```

Segmentation



Threshold = 50



Threshold = 75

Image Thresholding

- Automatic T**hresholding** : **Otsu Method**

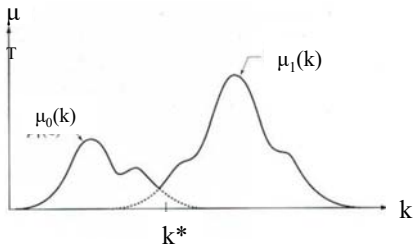


Figure 2: Gray-level PDFs of two regions in an image

$$\omega_0\mu_0 + \omega_1\mu_1 = \mu_T, \qquad \omega_0 + \omega_1 = 1.$$

$$\sigma_0^2 = \sum_{i=1}^{k} (i-\mu_0)^2 \Pr(i|C_0) = \sum_{i=1}^{k} (i-\mu_0)^2 p_i/\omega_0$$

$$\sigma_1^2 = \sum_{i=k+1}^{L} (i-\mu_1)^2 \Pr(i|C_1) = \sum_{i=k+1}^{L} (i-\mu_1)^2 p_i/\omega_1.$$

$$\sigma_B^2(k^*) = \max_{1 \le k < L} \sigma_B^2(k).$$





Nobuyuki Otsu. (1979). “A Threshold Selection Method from Gray-Level Histograms”. IEEE Transactions on Systems, MAN and Cybernetics, vol. SMC-9, No 1

Image Thresholding

- Global Fixed/ Global Automatic/
- Local Adaptive I/II threshoding methods
  : Adaptive Thresholding calculates a unique threshold $T(x, y)$ for each pixel based on its local neighborhood $b \times b$ block. This is essential for images with uneven illumination or shadows.

$$T(x,y) = \frac{1}{b^2}\sum_{i,j \in Block} f(i,j) - C$$



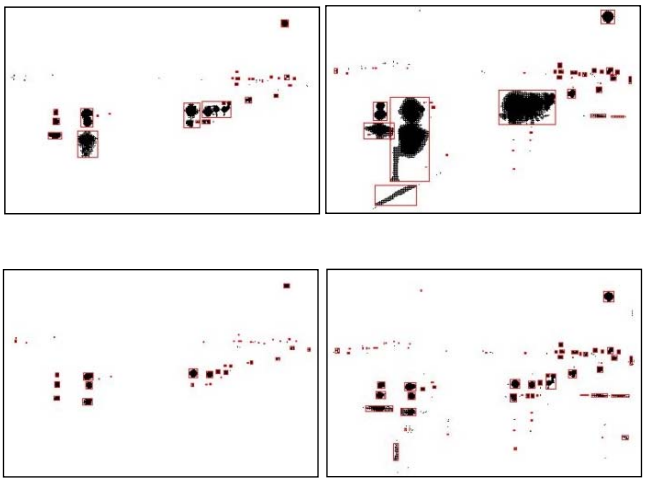
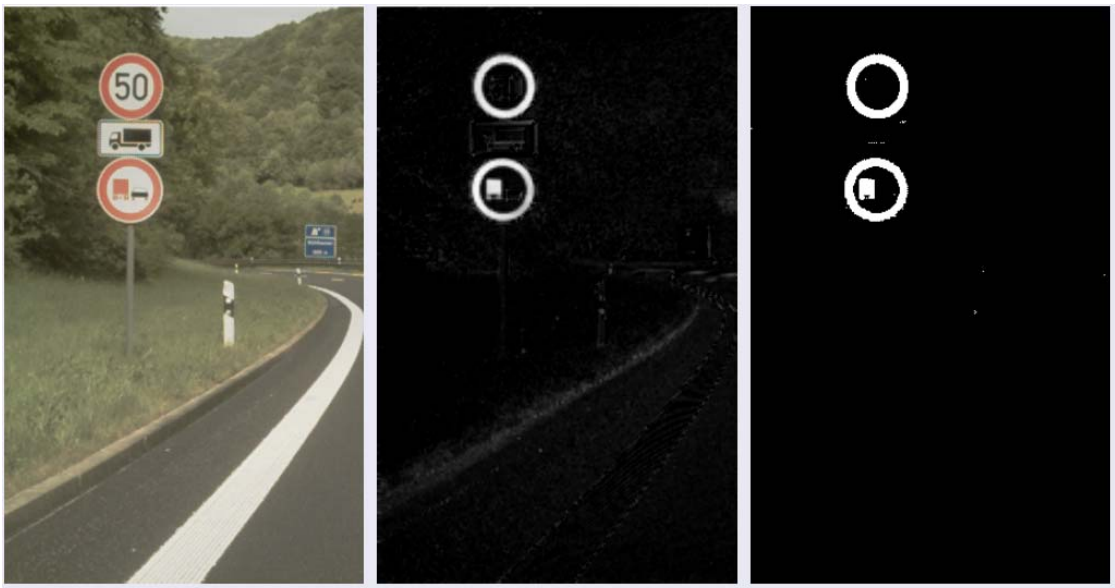


Image Thresholding

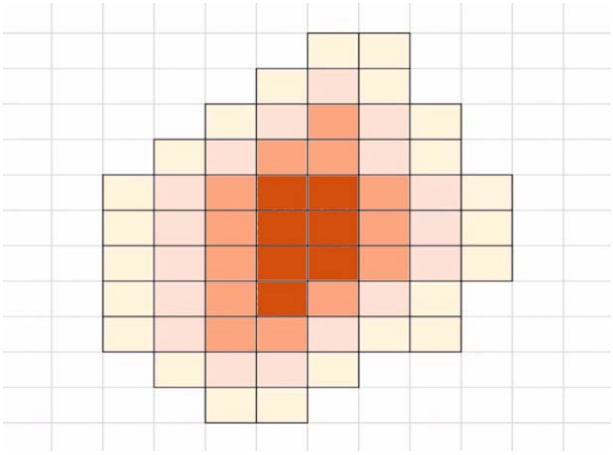
- Local Adaptive thresholding

$$f_R = \max(0, \frac{\min(R-G, R-B)}{S})$$
$$f_G = \max(0, \frac{\min(G-R, G-B)}{S})$$
$$f_B = \max(0, \frac{\min(B-R, B-R)}{S})$$
$$f_Y = \max(0, \frac{\min(R-B, G-B)}{S})$$
where $S = R+G+B$



(a) Original (b) Enhanced [Ruta10] (c) Thresholded

Thank You
Next : Binary Image Processing



Computer Vision
Report 04: Histogram Equalization

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주요내용

- RGB (원본)➔ YCbCr ➔ Y 정보만으로 Histogram Equalization ➔RGB(복원) 컬러 영상(복원)
- RGB (원본) 과 RGB 컬러 영상(복원) 의 PSNR 구하는 프로그램 작성

- 방법 1 : Histogram Equalization 을 수업 시간에 배운 아래수식 사용

$$h(i) = round\left(\frac{cdf(i) - cdf_{min}}{N - cdf_{min}} \times L\right)$$

- 방법 2: OpenCV 함수 사용

- 방법 1과 2 결과 비교

Report #4

- 구현한 프로그램의 코드와 실행 결과를 캡처하고 워드 또는 아래 한글에 붙여 넣어 파일을 강의 지원에 제출

- 파일명: report05_성명_학번.hwp

- 2 가지로 구현 하여 그 결과를 비교
  - 구현 1: OpenCV equalizeHist 함수를 이용하여 구현
  - 구현 2: 앞에서 주어진 수식을 이용하여 Histogram Equalization 구현
  - 결과 PSNR이 동일한지 확인하여 그 프로그램과 결과물을 제출
  - Y영상을 캡처하는 것이 아닌 Y정보만으로 Histogram Equalization한후 RGB 복원한 영상을 캡처
  - Time 모듈을 사용하여 OpenCV equalizeHist 함수 와 주어진 수식을 이용하여 직접 구현한 Histogram Equalization 의 실행시간 비교
  - Python 또는 C++ 중 하나만 구현
  - 주석 꼭 작성할 것
  - 프로그램 소스는 아래 한글에 폰트 크기 12이상으로 붙여 넣기하고 결과 영상은 캡쳐하여 넣어서 하나의 파일로 제출

Report #4 실행 결과 예)

